

Convolution of $e^{-|x|}$ and Fourier Representation

Source: J. E. Santos, Cambridge Part IB Complex Methods, Example Sheet 3, Question 2.

2. Show that the convolution of the function $e^{-|x|}$ with itself is given by $f(x) = (1 + |x|)e^{-|x|}$. Use the convolution theorem for Fourier transforms to show that

$$f(x) = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{e^{ikx}}{(1+k^2)^2} dk \quad (1.1)$$

and verify this result by contour integration.

Solution. The convolution of $g(x) = e^{-|x|}$ is given by

$$(g * g)(x) = \int dz e^{-|z|} e^{-|x-z|}. \quad (1.2)$$

For $x > 0$, the convolution becomes

$$(g * g)(x) = \int_x^{\infty} dz e^{-z} e^{-(z-x)} + \int_0^x dz e^{-z} e^{-(x-z)} + \int_{-\infty}^0 dz e^z e^{-(x-z)} = (1+x)e^{-x}. \quad (1.3)$$

Similarly, for $x < 0$, we find

$$(g * g)(x) = (1-x)e^x. \quad (1.4)$$

Hence, we find

$$(g * g)(x) = (1 + |x|)e^{-|x|}. \quad (1.5)$$

Using the Fourier transform of $e^{-|x|}$, the convolution can be written as

$$\begin{aligned} (g * g)(x) &= \frac{1}{\pi^2} \int dk \int dq \int dz \frac{e^{ikz} e^{iq(x-z)}}{(k^2+1)(q^2+1)} \\ &= \frac{2}{\pi} \int dk \int dq \frac{e^{iqx} \delta(k-q)}{(k^2+1)(q^2+1)} \\ &= \frac{2}{\pi} \int dk \frac{e^{ikx}}{(k^2+1)^2} = f(x). \end{aligned} \quad (1.6)$$

It is easy to see that the integrand has two poles of order 2 at $k = \pm i$. The corresponding residuals are given by

$$\begin{aligned} \text{Res} \left(\frac{e^{ikx}}{(k^2 + 1)^2}, k = \pm i \right) &= \lim_{k \rightarrow \pm i} \frac{d}{dk} \left(\frac{e^{ikx}}{(k \pm i)^2} \right) = \lim_{k \rightarrow \pm i} \frac{ixe^{ikx}(k \pm i) - 2e^{ikx}}{(k \pm i)^3} \\ &= \frac{1 \pm x}{4i} e^{\mp x}. \end{aligned} \quad (1.7)$$

For $x > 0$, we close the contour upward, while for $x < 0$ we close the contour downward. By simple algebra, we recover the result

$$(g * g)(x) = (1 + |x|)e^{-|x|}. \quad (1.8)$$